PART_A

Define the main goal of Analysis Phase?

T The main goal of the Analysis Phase is to understand business needs and define system requirements.
It focuses on studying the current system and identifying improvements.
Requirements are gathered, analyzed, and documented clearly.
This phase defines *what* the system should do, not *how* it will be built.

Identify at least 3 roles involved (e.g., Product Owner, Business Analyst, Project Manager)?

Business Analyst (BA): Gathers and documents requirements, bridging users and developers to meet business goals.

Product Owner: Represents end users, prioritizes features, and ensures the system delivers business value.

Project Manager (PM): Manages project scope, schedule, and communication to keep the analysis phase on track.

Other roles: System Analyst, Subject Matter Expert (SME), and End Users.

List 2 real tools or techniques used (e.g., JIRA, interviews, requirement workshops)?

Interviews (Technique): Used to gather information from stakeholders, understand their needs, and identify problems or improvements.

JIRA (Tool): A software tool for documenting, tracking, and managing requirements, tasks, and user stories.

Explain how missing requirements can affect the entire SDLC?

Missing requirements can cause serious problems in the SDLC:

1. **Design errors:** The system may be designed incorrectly, missing key features.
2. **Development delays:** Developers may need to rework or change mid-projects.
3. **Testing issues:** Test cases may not cover missing functionality, leading to defects.
4. **Higher costs:** Fixing issues later in the SDLC is more expensive than early correction.

5. **User dissatisfaction:** The final system may not meet business needs or expectations.

   Find one real-world example (from a company or project) that faced analysis-related issues?

FBI's Virtual Case File (VCF) project

- Goal: Modernize case management system.
- Problem: Poor analysis, unclear and changing requirements.
- Result: Cost $170M; project failed, system unusable.

  ✅ Shows how missing or unclear requirements can derail a project.

   Describe the main outputs (deliverables) of Design Phase?

   Main outputs of the Design Phase:

1. System Architecture: Blueprint hardware, software, and network structure.
2. Database & Data Models: Schemas, tables, and relationships.
3. User Interface Designs: Screens, layouts, and navigation flow.
4. Detailed Specifications: Functional and technical design documents.
5. Prototypes (optional): Mockups for validation before development.

   List 2–3 tools or techniques used for software/system design (e.g., UML, Figma, Lucidchart)?

   Tools/Techniques for Design Phase:

1. UML (Unified Modeling Language): Creates diagrams for system structure and behavior.
2. Figma: Designs user interfaces and prototypes.
3. Lucidchart: Draws flowcharts, ER diagrams, and architecture diagrams.

   Explain the difference between system design and UI/UX design? System Design: Focuses on the overall architecture, database, and technical components of a system. **UI/UX Design:** Focuses on user interface, experience, and how users interact with the system. ✅ **In short:** System design is about *how the system works*, UI/UX design is about *how it feels to the user*.

   Describe one example where poor design decisions increased project cost or failure risk?

   Example: Denver International Airport Baggage Handling System (1990s)
   Issue: Complex automated baggage system was poorly designed and over-ambitious. Result**:** Massive delays, redesigns, and cost overruns of over $500 million.

✅ Shows how poor design decisions can greatly increase cost and project risk.

Describe what happens in the Development Phase?

Developers write the actual code based on design specifications.
Databases, interfaces, and system components are built and integrated.
Unit testing is done to ensure that each part works correctly.

✅ **In short:** This phase turns designs into a working software system

Identify front-end and back-end responsibilities?

Front-End	Back-End
Design and implement UI	Develop server, database, and logic
Ensure good UX and responsiveness	Manage data storage, retrieval, and security
Handle client-side interactions	Handle business rules and server-side processing

List 3 tools or technologies commonly used by developers (e.g., GitHub, VS Code, Spring Boot)?

Common Developer Tools/Technologies:

GitHub – Version control and collaboration.
VS Code – Code editor for writing and debugging.
Spring Boot – Framework for building back-end applications.

Explain why proper version control is essential in this phase?

Proper version control is essential because:

- It tracks changes in code, preventing loss of work.
- Allows multiple developers to work together without conflicts.
- Makes it easy to revert to previous versions if bugs or issues occur. ✅ **In short:** Ensures code integrity, collaboration, and safer development.

Describe a real case of a software project that failed due to poor development practices?

Example: Healthcare.gov Launch (2013)
Issue: Poor development practices, lack of integration testing, and last-minute coding.
Result: Website crashed at launch, many users couldn't sign up, and costly fixes were needed.

✅ Shows how bad development practices can cause major project failure.

Explain the purpose of Testing Phase?

Verify that the software works as intended and meets requirements.

Identify and fix bugs or defects before release.
Ensure reliability, performance, and user satisfaction.

Identify 3 types of testing (e.g., unit, integration, system)?

Types of Testing:

Unit Testing: Checks individual components for correctness.

Integration Testing: Ensures combined modules work together.

System Testing: Validates the complete system against requirements.

List 2–3 testing tools (e.g., Selenium, JMeter, Postman)?

Selenium: Automates web application testing.

JMeter: Tests performance and load.

Postman: Tests APIs for functionality.

Explain how automated testing improves efficiency?

Automated Testing Benefits:

Run tests faster and more frequently.
Reduces human errors in repetitive tasks.
Allows early detection of bugs and easier regression testing

Find one case study or article showing how testing caught critical bugs before?

NASA's Mars Climate Orbiter faced software errors.
Testing identified unit conversion mistakes in simulations.
Early detection prevented critical mission failure.
Shows the importance of thorough testing before release.

Define what deployment means in SDLC?

Deployment is releasing the software to production so users can access it.
It ensures the system is installed, configured, and running correctly.
Includes moving code from development to live environment.
Ensures that the system is ready for real-world use.

Mention 2 deployment models (e.g., manual vs. Continuous?

Manual Deployment: Developers deploy updates step by step.

Continuous Deployment: Automated release of code after testing.

Speeds up delivery and reduces human error.
Supports agile and DevOps practices.

Identify 3 tools or platforms used in deployment (e.g., Jenkins, AWS, Docker)?

Jenkins: Automates build and deployment processes.

AWS: Provides cloud infrastructure for deployment.

Docker: Packages applications for consistent deployment.
Ensures faster and reliable releases.

Explain how DevOps practices support this phase?

Integrates development and operations for smoother deployment.
Automates testing, build, and release pipelines.
Improves collaboration and reduces downtime.
Enable faster delivery and consistent deployments

Describe a real incident where deployment issues affect users (e.g., app crash or downtime)?

Example: Twitter outage due to a deployment bug.
A new update caused server failures.
Millions of users experienced downtime.
Highlights the need for careful deployment and monitoring.

Define why Maintenance Phase never truly ends?

Software requires ongoing updates, bug fixes, and improvements.
New features, security patches, and performance tuning continue.
User needs and environments change over time.
Maintenance ensures that the system remains functional and relevant.

List 3 maintenance activities (e.g., bug fixing, updating libraries, performance tuning)?

Bug Fixing: Correcting errors reported by users.
Updating Libraries: Ensuring software uses supported dependencies.
Performance Tuning: Improving speed, scalability, and reliability.
Keeps the system stable and efficient.

Identify 2 tools used for tracking and maintaining systems (e.g., Jira, ServiceNow)?

Jira: Tracks bugs, tasks, and progress.
ServiceNow: Manages IT incidents and system maintenance.
Supports efficient workflow and accountability.

Explain how user feedback influences maintenance decisions?

Users report bugs or suggest improvements.
Feedback helps prioritize updates and feature enhancements.
Guides for maintenance planning and decision-making.
Ensures the system meets real-world needs.

Describe a case where regular maintenance prevents major system failure?

Example: Microsoft Windows Updates.

Regular patching fixes security vulnerabilities and bugs.
Prevents system crashes and malware exploits.
Ensures millions of users have a stable and secure OS.

PART_B

SDLC Phase	How it was applied in WhatsApp
Analysis	The team identified user needs for instant messaging, multimedia sharing, group chats, and cross-platform access. They focused on simplicity, speed, and privacy.
Design	UI/UX was designed for a clean chat interface, easy navigation, and minimal clicks. System design focused on efficient message delivery and end-to-end encryption.
Development	Built using technologies like Erlang for backend messaging, Java/Kotlin for Android, Swift for iOS, and databases optimized for real-time messaging.
Testing	Unit, integration, and system testing ensured messages were delivered reliably. Beta testing with real users helped catch edge-case bugs.
Deployment	Updates are rolled out via app stores using phased releases. Continuous deployment allows rapid bug fixes and feature improvements.
Maintenance	Bugs are tracked via internal tools, feedback from users is collected, and features like status updates or voice/video calls are added regularly.